BUILDING ENERGY CARD · ARCHETYPE-QC

R_SF_Det_1fl_1985-2012

VERSION 2026-05

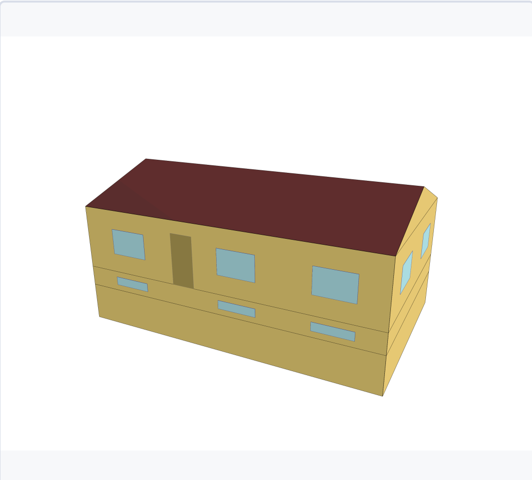
DATE 2026-06-01

SECTION 1 — MODEL DESCRIPTION

BUILDING DESCRIPTION

Sector	Residential
Building Type	Single-Family
Building Sub Type	Detached
Building Size	1 Floor
Vintage	1985-2012

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation and performance analysis.

BUILDING CHARACTERISTICS

Floor Area	180.0 m²
Climate Zone	6A
HVAC SYSTEM	
cooling system	Air-Source Heat Pump (Mini-Split)
Heating system	Electric Baseboards
Ventilation system	/
Typical COP cooling	2.5

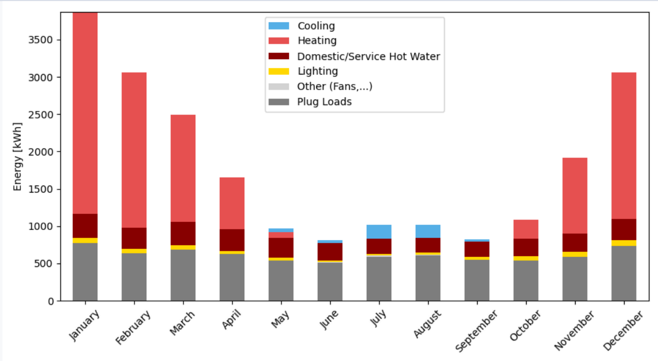
ENVELOPE

INSULATION & AIR TIGHTNESS	
Walls Ext [W/m2-K]	0.48
Walls Int [W/m2-K]	0
Roof [W/m2-K]	2.86
Slabs [W/m2-K]	0
GLAZING	
Glazing [W/m2-K]	2.94
SHGC	0.6
WWR (WINDOW-TO-WALL RATIO)	
Average	13.02

ANNUAL END USES — ELECTRICITY [KWH]

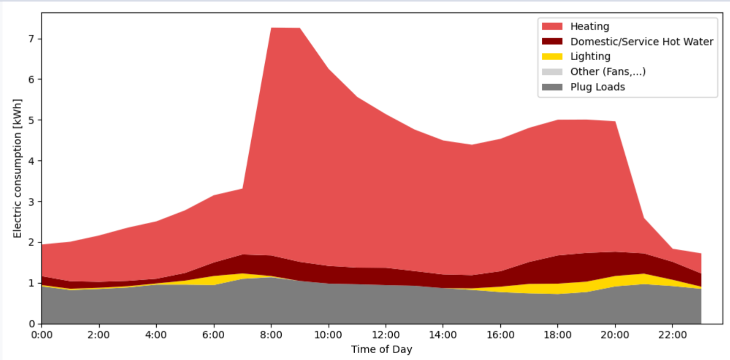
END USE	KWH/YR
Heating	10241.7
Cooling	483.3
Interior Lighting	444.4
Exterior Lighting	122.2
Interior Equipment	10455.6
Fans	36.1
Total End Uses	21786.3

MONTHLY CONSUMPTION BY END USE



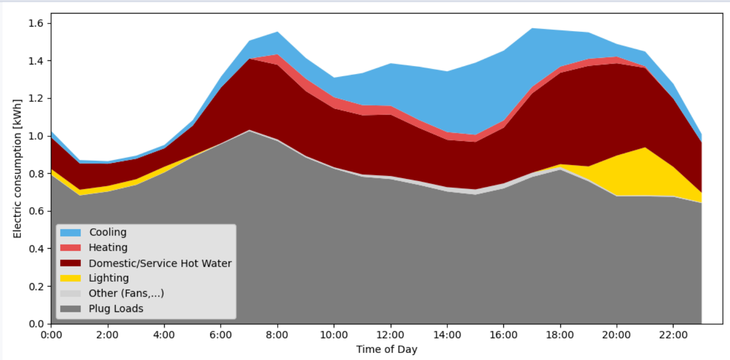
Monthly energy consumption breakdown by end use, showing the contribution of heating, cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout the year.

TYPICAL DAILY CONSUMPTION - WINTER



Stacked daily winter energy consumption profile showing the contribution of heating, lighting, domestic hot water, plug loads, and auxiliary systems throughout a typical winter day.

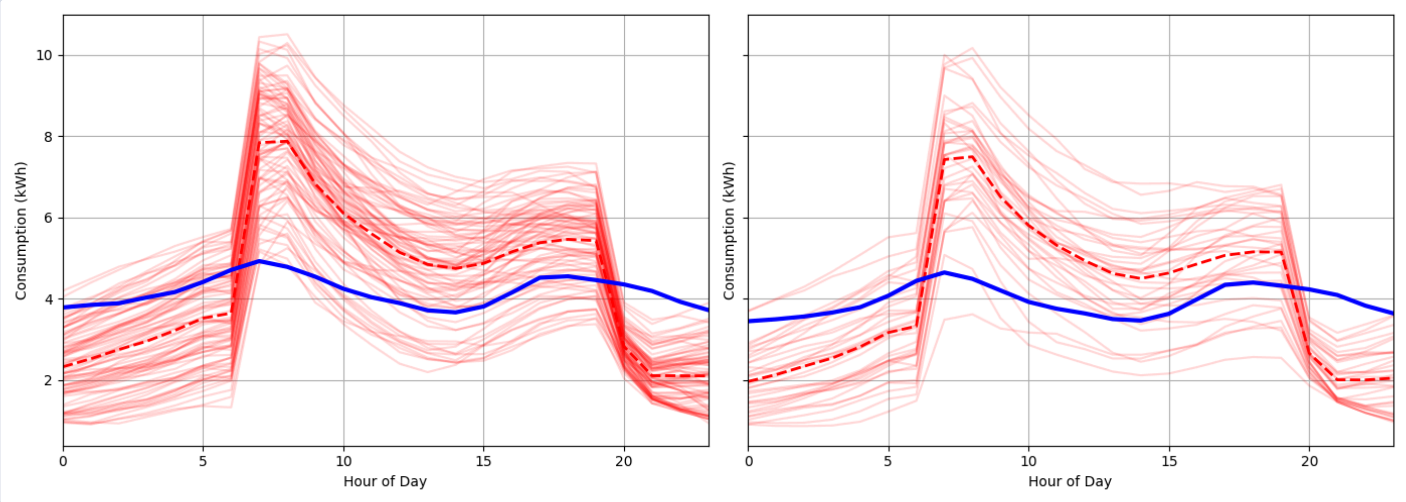
TYPICAL DAILY CONSUMPTION - SUMMER



Stacked daily summer energy consumption profile showing the contribution of cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout a typical summer day.

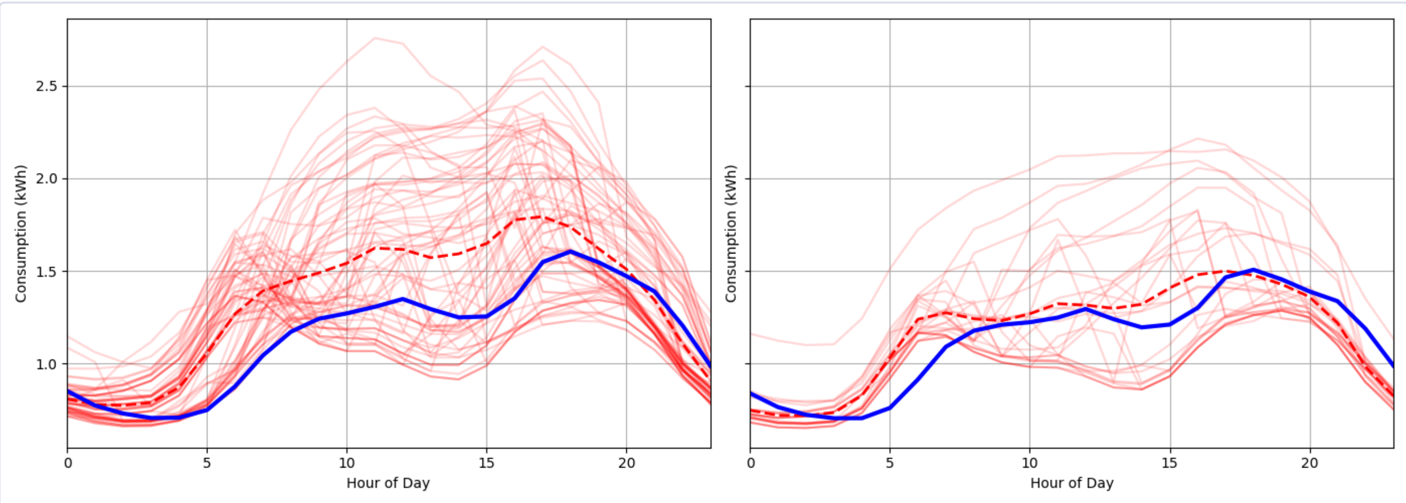
ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS
80.7 kWh/m²/yr	21786.0 kWh/yr	21786.0 kWh/yr	0 kWh/yr

TYPICAL DAY — WINTER



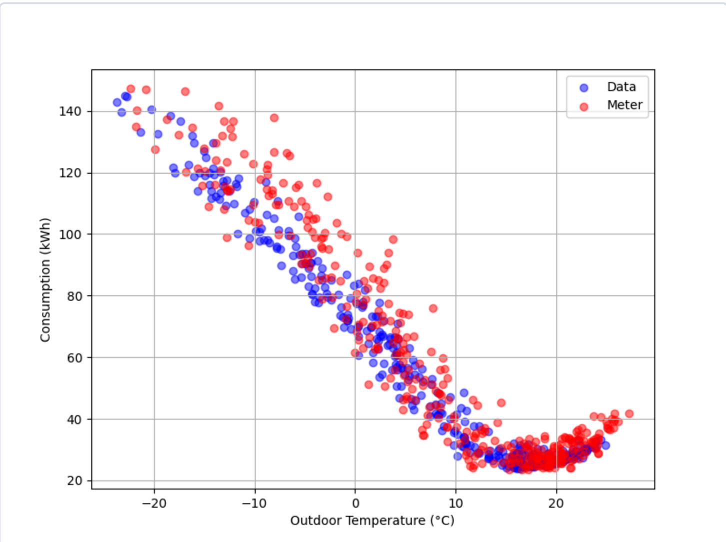
Winter daily consumption profile comparing simulated and measured hourly energy use. Red lines represent individual winter days from the model, the red dashed line indicates the average simulated profile, and the blue line shows measured data for comparison. The left plot represents weekdays, while the right plot represents weekends.

TYPICAL DAY — SUMMER



Summer daily consumption profile comparing simulated and measured hourly energy use. Red lines represent individual winter days from the model, the red dashed line indicates the average simulated profile, and the blue line shows measured data for comparison. The left plot represents weekdays, while the right plot represents weekends.

PRISM CURVE



PRISM curve comparing simulated (OPE) and measured energy consumption as a function of outdoor temperature, illustrating the relationship between weather conditions and building energy demand across the year.

COMPARISON TABLE — KEY INDICATORS

INDICATOR	MODEL	REFERENCE	DELTA
Yearly electricity consumption (kWh)	21785.52	20983.84	801.7
Yearly baseload consumption (kWh/jour)	27.86	26.63	1.2
Slope electricity heating (W/K)	-147.33	-136.0	-11.3
Slope climatisation cooling (W/K)	74.92	52.56	22.4
Peak winter AM (kW)	10.5	7.06	3.4
Hour peak winter AM	8.0	7.0	1.0
Peak winter PM (kW)	7.61	6.51	1.1
Hour peak winter PM	12.0	17.0	-5.0